(Please write your Enrollment Number) Enrollment No. 01601172024

MID-TERM EXAMINATION
(Course Name _B.Tech IT/ CSE-AI/ CSE) (Semester 5th)
(October, 2023) OFF LINE mode

Subject Code: BCS 303	Subject: Theory of Computation
Time : 1 ½Hours	Maximum Marks : 30

Note: Q. 1 is compulsory.

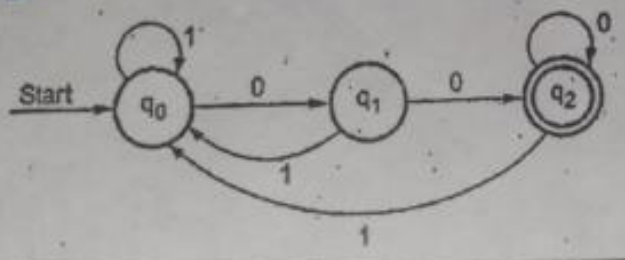
Q1 (2.5*4)

(a) Define Finite Automata and Discuss the difference between DFA and NDFA?

(b) How many tuples are used to define Grammar? Explain the tuples of grammar with the help of an example grammar.

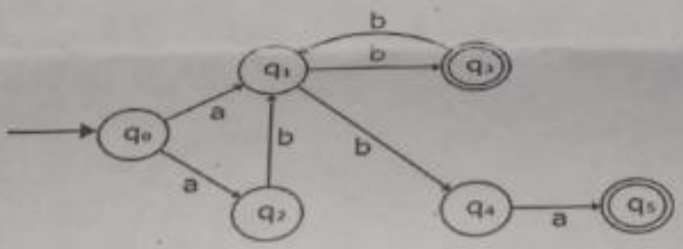
(c) What are Null Production, Useless Symbols, and Unit Production in grammar? Discuss with example.

(d) Find the language accepted by the following Finite Automata?



Q2 (Attempt any Two Parts) UNIT-1 (5,5)

(a) Convert the following NFA to DFA.



(b) Design a DFA that accepts all the strings not having more than two a's over the alphabet $\Sigma = \{a, b\}$. And check whether the string **abababaaabbbba** is accepted.

(c) Define Pumping Lemma for Regular Languages. Explain with a suitable example.

Q3 (Attempt any Two Parts) **UNIT-2** (5,5)

(a) What is ambiguity? Find the equivalent unambiguous grammar for the following grammar:
E→E+E
E→E*E
E→id

(b) What is CNF? Convert the following CFG to CNF?
S→ABa
A→aab
B→Ac

(c) Simplify the following CFG? Discuss the simplification procedure.
S→aSSS/aA/aBB
A→aAA/B/ϵ
B→bB/bbC
C→B
D→a/bbc